

GhostNet — Project Status

Multi-Agent Ghost Net & Marine Debris Intelligence Pipeline
6 September 2026 | commit e2fa0e9

1. Acceptance criteria (PRD §12)

Two of six fully met; a third became half-met today.

Criterion	Status	Note
Six-agent pipeline end to end on a real region, producing a ranked plan	PARTIAL	Ran on real data for the first time today. 3 of 6 agents on real inputs, 3 degraded for want of data.
Detection + verification benchmarked on a published dataset	DONE	Precision 0.238 → 0.623 , F1 0.385 → 0.753 on held-out MARIDA.
Drift trajectories vs NOAA drifter paths, with an error metric	NOT DONE	Unblocked today. Needs a harness script, ~2 h.
Attribution vs Ocean Cleanup; dark vessel vs GFW	NOT DONE	Needs one download and one unwritten client.
Ablation: each agent's removal degrades output explainably	PARTIAL	Verification ablation measured on real data; system-level ablation only on synthetic.
Every output traceable to its underlying evidence	DONE	Enforced by an automated check, not just documented.

2. Agents

FR	Agent	Status	Evidence
FR-1.1	Sentinel-2 ingestion	DONE	Streams from a public catalogue; no credential.
FR-1.2/1.3	Spectral-index detection	DONE	Region recall 0.407 on MARIDA test.
FR-1.4	CNN detector	DONE	Region recall 0.703 ; 7.6× fewer candidates.
FR-2.1/2.3	False-positive checks + reasons	DONE	Five checks, per-detection reason text.
FR-2.2	Multi-temporal consistency	DONE	Measured negative — see section 4.
FR-2.4	Before/after precision-recall	DONE	The project's headline result.
FR-3	Drift, backward/forward + envelope	DONE	Code complete. Its <i>validation</i> is criterion 3.
FR-4	Source attribution	PARTIAL	Code and tests done; no river table yet.
FR-5	Dark-vessel correlation	NOT DONE	Client not implemented.
FR-6.1	Priority score	PARTIAL	Runs; ecological-risk component dropped without MPA data.
FR-6.2–6.4	Capacity plan, rationale, approval	DONE	Including the named-reviewer gate.

3. What is left, in priority order

Both API tokens now exist and are verified — credentials are no longer the bottleneck. Two downloads, one unwritten client and the report are.

#	Task	Owner	Blocked on	Effort
1	Download Protected Planet (WDPA marine), run the extract command	Ashwin	a web form	30 min
2	Download the Meijer 2021 river table, run the extract command	Ashwin	a download	30 min
3	Write the drift-validation harness — closes criterion 3	A	nothing	~2 h
4	Implement the GFW SAR client — closes FR-5	B	nothing	~4 h
5	Re-export a full real run once 1–4 land	Ashwin	1, 2, 4	1 h
6	System-level ablation on real inputs	Ashwin	5	1 h
7	Write the report (outline exists, prose does not)	all	nothing	days
8	Deploy the console to a URL	Ashwin	an account	30 min

4. Three results that must be quoted correctly

These are where an examination goes wrong, so they are stated here rather than left in the appendix.

1. The verification gain and region recall are always quoted together. A precision gain of +0.385 on its own implies the system finds nearly all the debris. It lands on 96 of 236 annotated regions. An automated test enforces the pairing.

2. Region recall 0.703 is a within-tile number. The benchmark splits by patch, so 91% of test patches sit on tiles the model trained on. On a genuinely unseen region the cost is -0.072 F1, almost all of it recall.

3. Multi-temporal consistency (FR-2.2) is a measured negative, not a contribution. Given a real current field its harm disappears (six false rejections become none) but its contribution is still exactly zero, and the cause is now named: nearest-neighbour matching means its strongest signal never fires. This is recorded as a measured exception to the design test that every agent is load-bearing.

5. Data and machines

Dataset	Present	Needed for
MARIDA benchmark	DONE	Benchmarking
OSCAR surface currents	DONE	Drift modelling
Drifter tracks	DONE	Drift validation
River emissions table	NOT DONE	FR-4 attribution
Protected Planet (MPA)	NOT DONE	FR-6.1 ecological risk
Global Fishing Watch	NOT DONE	FR-5 dark vessels
Sentinel-2 imagery	STREAMED	No local copy needed

The work is split across two machines: a workstation with the GPU, the benchmark and model training, and a laptop for agent code, the console and documentation. Neither machine may assume the other's files are present. Git is the source of truth, not chat history.

6. The rule that governs the project

“No number reaches the results file that the person who produced it cannot derive and defend.”

An examiner asking “why is region recall 0.703?” is asking the person, not the tool.